

Mathematical Induction

→ Mathematical induction is an important proof techniques which can be used to prove mathematical theorems or statement and show the correctness of the computer program.

principle of mathematical Induction

→ To prove $p(n)$ is true for all positive integers 'n', where $p(n)$ is a propositional function, we complete two steps

Basis step: verify that $p(n)$ is true for $n = 0$ or 1 .

Inductive step: Show that the conditional statement $p(k) \rightarrow p(k+1)$ is true for all positive integers 'k'.

Example:1 Show that if n is a positive integer, then

$$1+2+\dots+n = \frac{n(n+1)}{2}.$$

Soln

Let $p(n)$ be the proposition that the sum of first 'n' positive integer is $\frac{n(n+1)}{2}$

To prove that $p(n)$ is true, we have to show $p(1)$ is true and assuming that $p(k)$ is true prove that $p(k+1)$ is true.

Basis step: $p(1)$ is true because $p(1) = 1 = \frac{1(1+1)}{2} = \frac{2}{2} = 1$.

Inductive step: Assume that $p(k)$ is true.

$$p(k) = 1+2+\dots+k = \frac{k(k+1)}{2}, \quad \text{--- eqn (1)}$$

To show $p(k+1)$ is true,

Adding $(k+1)$ on both sides of the equation (1)

$$\begin{aligned} 1+2+\dots+k+(k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}, \text{ which is true.} \end{aligned}$$

This shows that $P(k+1)$ is also true.

Hence, proved.

Example 2: Use mathematical induction to show that

$$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1 \text{ for all non-negative integers 'n'..}$$

Soln

Let $P(n)$ be the proposition that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for integer n .

Basis Step: $P(0)$ is true because $1 = 2^{0+1} - 1 = 2 - 1 = 1$.

Inductive step: Assume that $P(k)$ is true.

$$1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1 \quad \text{--- (1)}$$

To show $P(k+1)$ is true, add 2^{k+1} to both side of eqⁿ (1)

$$\begin{aligned} 1 + 2 + 2^2 + \dots + 2^k + 2^{k+1} &= 2^{k+1} - 1 + 2^{k+1} \\ &= \cancel{2 \cdot 2^{k+1}} = 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+2} - 1 \end{aligned}$$

Which is true.

Hence proved.

Example: 3. Conjecture a formula for sum of the first 'n' positive odd integers. Then prove your conjecture using mathematical induction.

Soln

$$\begin{array}{lll} 1 = 1 & 1 + 3 + 5 = 9 & 1 + 3 + 5 + 7 + 9 = 25 \\ 1 + 3 = 4 & 1 + 3 + 5 + 7 = 16, & \text{and so on.} \end{array}$$

So,

$$1 + 3 + 5 + \dots + (2n-1) = n^2.$$

Now, to prove above conjecture.

Basis step: $P(1) = 1 = 1^2 = 1$
which is true.

$$\begin{aligned} \text{For } k+1 & \Rightarrow 2(k+1)-1 \\ & = 2k+2-1 \\ & = 2k+1 \end{aligned}$$

Inductive step: Let $p(k)$ is true
 $1+3+5+\dots+(2k-1) = k^2$ — eqⁿ ①

Now, adding $(2k+1)$ to both side of the equation ① we get,

$$\begin{aligned} 1+3+5+\dots+(2k-1) + (2k+1) &= k^2 + 2k + 1 \\ &= (k+1)^2 \end{aligned}$$

which is true.

hence, proved.

Example: 4. Use Mathematical Induction to show that the sum of finite number of terms of a geometric progression

$$\sum_{j=0}^n ar^j = a + ar + ar^2 + \dots + ar^n = \frac{ar^{n+1} - a}{r-1} \quad \text{When } r \neq 1.$$

Where n is non-negative integer.

Solⁿ

Let $p(n)$ be the propositional statement.

$$p(n) = a + ar + ar^2 + \dots + ar^n = \frac{ar^{n+1} - a}{r-1} \quad \text{when } r \neq 1 \text{ \& } n \text{ is non-negative integer.}$$

Basis step: $p(0)$ is true because $p(0) = a = \frac{ar^1 - a}{r-1} = \frac{a(r-1)}{(r-1)} = a.$

which is true.

Inductive step: Assume $p(k)$ is true.

$$p(k) = a + ar + ar^2 + \dots + ar^k = \frac{ar^{k+1} - a}{r-1} \quad \text{--- eqⁿ ①}$$

Adding ar^{k+1} to both side of the equation ①, we get

$$\begin{aligned}
 a + ar + ar^2 + \dots + ar^k + ar^{k+1} &= \frac{ar^{k+1} - a}{r-1} + ar^{k+1} \\
 &= \frac{ar^{k+1} - a + ar^{k+1}(r-1)}{(r-1)} \\
 &= \frac{\cancel{ar^{k+1}} - a + ar^{k+2} - \cancel{ar^{k+1}}}{(r-1)} \\
 &= \frac{ar^{k+2} - a}{r-1}
 \end{aligned}$$

Which is true, hence proved

Proving Inequalities

* Use mathematical induction to prove the inequality $n < 2^n$ for all positive integers $n \geq 1$.

Soln Let $p(n)$ be the proposition that $n < 2^n$ for all positive integers n .

Basis Step: $p(1)$ is true because $1 < 2^1$ which is true.

Inductive Step: Assume $p(k)$ is true.

$$p(k) = k < 2^k \quad \text{--- (1)}$$

~~Adding 1 to both side of equation (1)~~

$$k + k + 1 < 2^k + (k + 1)$$

$$k + 1 < 2^k + 1 < 2^k +$$

To show that $p(k+1)$ is true ~~let us~~ add 1 to both side

$$\begin{aligned}
 k+1 &< 2^{k+1} \\
 &< 2^k \cdot 2 \\
 &\leq 2
 \end{aligned}$$

$$\begin{aligned}
 k+1 &< 2^k + 1 \\
 &\leq 2^k + 2^k \quad [1 \leq 2^k] \\
 &\leq 2 \cdot 2^k \\
 &= 2^{k+1}
 \end{aligned}$$

* Use mathematical induction to show that $n^3 - n$ is divisible by 3 whenever n is a positive integer.

Soln Let $p(n)$ be the proposition that $n^3 - n$ is divisible by 3.

Basis step: $p(1)$ is true since $1^3 - 1 = 0$ is divisible by 3.

Inductive Step:

Assume that $p(k)$ is true

$\therefore p(k) = k^3 - k$ is divisible by 3.

To show that $p(k+1)$ is true put $n = k+1$.

$$\begin{aligned} \text{So, } p(k+1) &= (k+1)^3 - (k+1) \\ &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= (k^3 - k) + 3(k^2 + k) \end{aligned}$$

Since both terms in this sum are divisible by 3. It follows that

$(k^3 - k) + 3(k^2 + k)$ is divisible by 3.

hence, proved

H.W. * proved that 3 divides $n^3 + 2n$ whenever 'n' is a non-negative integer.

* Use mathematical induction to prove that $2^n < n!$ for every positive integer n with $n \geq 4$.

Soln Let $p(n)$ be the proposition that $2^n < n!$

Basis step: To prove the inequality for $n \geq 4$ the basic step is

$p(4)$. $p(4)$ is true since $2^4 < 4! \Rightarrow 16 < 24$ which is true.

Inductive step: Assume that $p(k)$ is true for the integer k with $k \geq 4$.

i.e. ~~$p(k)$~~ $2^k < k!$ for $k \geq 4$.

We have to show that $p(k+1)$ true

$$2^{k+1} = 2^k \cdot 2$$

$< 2 \cdot K!$ (by the inductive step $2^K < K!$)

$< (K+1)K!$ because $2 < K+1$ for $K \geq 4$.

$= (K+1)!$ by the definition of factorial function.

This shows that $P(K+1)$ is true when $P(K)$ is true.
Hence proved.

Strong Induction

→ The strong induction uses the same concept as mathematical induction method of proof but only difference between these two types of induction is in the inductive step.

→ The formal definition of strong induction is:

Let $P(n)$ be the proposition, to show that $P(n)$ is true for all positive integers 'n' we have to complete two steps.

Basis Step: Show that $P(n)$ is true for $n=0$ or 1.

Inductive Step: It is shown that $[P(1) \wedge P(2) \wedge \dots \wedge P(K)] \rightarrow P(K+1)$ is true for every positive integer 'K'.

Example

⊗ Use mathematical induction to show that 3 divides $n^3 + 2n$ whenever 'n' is a non-negative integer.

Soln

Assume that $P(n)$ is a proposition.

$P(n)$: $n^3 + 2n$ is divisible by 3.

Basis step: ~~show~~ $P(0)$ is true since $P(0) = 0^3 + 2 \times 0 = 0$ is divisible by 3.

Inductive step:

$P(1)$ is true since $1^3 + 2 \times 1 = 3$ is divisible by 3.

$P(2)$ " " " $2^3 + 2 \times 2 = 12$ is divisible by 3.

$P(3)$ " " " $3^3 + 2 \times 3 = 15$ is divisible by 3.

$P(k)$ is " " " $k^3 + 2k$ is divisible by 3.

$P(k+1)$: $(k+1)^3 + 2(k+1)$

$$= k^3 + 3k^2 + 3k + 1 + 2k + 2$$

$$= k^3 + 2k + 3k^2 + 3k + 3$$

$$= (k^3 + 2k) + 3(k^2 + k + 1)$$

Since $(k^3 + 2k)$ is divisible by 3 and multiple of 3 is also divisible by 3.
 it follows $(k^3 + 2k) + 3(k^2 + k + 1)$ is also divisible by 3.

hence, proved

Well-ordering property

→ The well-ordering property states that

"Every non-empty set of non-negative integers has a least element".
 → Using this property, we verify the validity of proofs using mathematical induction.

Using mathematical induction, we prove $P(1)$ is true and $P(n) \rightarrow P(n+1)$ is true for all positive integers 'n'. If the proof by mathematical induction is not valid then $P(n)$ is true for all positive integers 'n' would be false. Let, the set of positive integers for which $P(n)$ is ~~not~~ false be T. Then T is non-empty since there is at least one element in T such that $P(n)$ is false.

By the well-ordering property, T has a least element, let the least element be K. We know that 'K' cannot be 1 because we have already proved $P(1)$ is true. So K is positive integer greater than 1 so $K-1$ is positive integer. So, we have $P(K-1)$ must be true. Here $K-1$ is less than K. i.e. $K-1$ is not in the set T. Since the implication $P(K-1) \rightarrow P(K)$ is true, $P(K)$ must be true. This contradicts the choice of K. Hence, $P(n)$ must be true for all positive integers 'n'.

Example: prove that $a^n = 1$ for all non-negative integers 'n', whenever a is a non-zero real number.

Soln Basis step: For $n=0$, $a^0 = 1$ which is true.

Inductive hyp step: Assume that $a^k = 1$ is true for all non-negative integers $1 \leq k \leq n$.

$$a^{k+1} = a^k \cdot a^k / a^{k-1} = 1 \cdot 1 / 1 = 1$$

hence, proved

Recursive Definitions

→ The method of defining a function in terms of itself is known as recursive definition of function.

→ Let us suppose that the domain of function is set of non-negative numbers, then we use two steps to define a function.

- ① Basis Step: Specify the value of function at 0.
- ② Recursive Step: Give a rule for finding its value at an integer from its value at smaller integers. such a definition is called recursive definition.

Example:

- ① Let the given sequence is 2, 5, 8, 11, 14, 17, ...

Solⁿ Then the n th term of this sequence can be defined as,

$$t_n = a + (n-1)d, \quad n \geq 1$$

$$t_n = 2 + (n-1)3 = 2 + 3n - 3 = 3n - 1$$

and the same sequence can be defined recursively as,

- ① Basis Step: $S(0) = 2$.

- ② Recursive Step: $S(n+1) = S(n) + 3, \quad n \geq 0$.

- ② The sequence 1, 3, 9, 27, ... can be defined recursively as,

- ① Basis Step: $S(0) = 1$.

- ② Recursive Step: $S(n+1) = 3S(n)$ for $n \geq 0$.

- ③ Let function f is defined as

$$F(0) = 2$$

$$F(n+1) = 2F(n) + 1$$

Then find the value of $f(1)$, $f(2)$ and $f(3)$.

Solⁿ $F(1) = 2f(0) + 1 = 2 \times 2 + 1 = 5$

$F(2) = 2 \times f(1) + 1 = 2 \times 5 + 1 = 11$

$F(3) = 2 \times f(2) + 1 = 2 \times 11 + 1 = 23$

⊛ Give a recursive definition of sequence $\{a_n\}$, $n = 1, 2, \dots, n$ if $a_n = 10^n$.

Solⁿ Basis step: $a_1 = 10^1 = 10$.

Recursive step: $a_n = 10 \times a_{n-1}$

Recursive Algorithm.

→ An algorithm is called recursive if it solves a problem by reducing it into an instance of the same problem with smaller input.

Example

① An algorithm to find the factorial of a given number.

```
int fact(int n)
{
    if (n == 0)
        return 1;
    else
        return n * fact(n-1);
}
```

② Recursive algorithm for computing a^n .

```
int power(int n, int a)
{
    if (n == 0)
        return a;
    else
        return a * power(a, n-1);
}
```

③ Recursive algorithm for computing gcd. (Greatest Common Divisor) with $a \leq b$.

```
int gcd(int a, int b)
{
    if (a == 0)
        return gcd(a, b) = b;
    else
        return gcd(a, b) = gcd(b % a, a);
}
```

④ Recursive algorithm for binary search algorithm.

⑤ Recursive algorithm for merge sort.

proving correctness of Recursive Algorithm

→ Mathematical induction and its variant, Strong Induction can be used to prove that a recursive algorithm is correct, that is, it produces the desired output for all possible input values.

Example: prove that recursive algorithm for computing a^n is correct,

ie. int power (int a, int n)

{

if (n == 0)

return 1;

else

return a * power(a, n-1);

}

Soln

Basis step: If $n=0$ then ~~a^0~~ $\text{power}(a, 0) = 1$. This is correct because " $a^0 = 1$ " for every non-zero real number a .

Inductive step:

Let $P(k)$ is true

So, $\text{power}(a, k) = a^k$ for all $a \neq 0$ for non-negative integer k .

We have to show that $P(k+1)$ is true.

So, For a^{k+1} ,

$$\begin{aligned} \text{power}(a, k+1) &= a \cdot \text{power}(a, k) \quad [\because a \times \text{power}(a, k+1-1) \\ &= a \times \text{power}(a, k) \\ &= a \cdot a^k \quad [\because \text{power}(a, k) = a^k] \\ &= a^{k+1} \end{aligned}$$

This is true.

Hence proved.

Recursive definition of a set

→ A definition of set that specifies an initial set of elements and in the set and a rule for obtaining other elements from those in the set.

Eg: If $\Sigma = \{0, 1\}$, the strings found to be in Σ^* is the set of all bitstrings specified to be in Σ^* . In the basis step, 0 and 1 are

formed during the first application of the recursive step.

→ The strings 00, 01, 10 and 11 are formed during the 2nd application of the recursive steps and so on.

Structural Induction

→ To prove recursively defined sets, we use a form of mathematical induction called Structural Induction.

→ The structural induction consists of two parts

(a) Basis step: Show that the result hold true for all elements specified in the basis step of the recursive definition to be in the set.

(b) Recursive step: Show that if the statement is true for each of the elements used to construct new elements in the recursive step of the definition, the result holds for those new elements.

Validity of structural Induction

→ The validity of ~~mat~~ structural induction can be seen as the validity of the mathematical induction.

→ If $P(n)$ denotes the statement that is recursively defined, for all positive integers 'n'.

The basis step of the structural induction method corresponds to the basis step of the mathematical induction method.

We can see the recursive step in the structural induction tells if $P(k)$ is true it implies $P(k+1)$ is true. Where $P(k)$ is assumed already and $P(k+1)$ is derived in terms of ~~$P(k+1)$~~ $P(k)$.

Hence, it follows the proofs by mathematical induction.